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Professional Summary

Electrical Engineering student at Toronto Metropolitan University focused on hardware design, digital design verification, and embedded systems. Designs and verifies RTL and analog circuits, brings up and debugs embedded hardware, and validates designs against fixed specifications using independent Python reference models, cocotb, SPICE, and bench instrumentation (oscilloscope, multimeter, function generator) alongside Quartus II, ModelSim, and Multisim. Seeks hardware and verification co-op roles involving RTL design, test automation, qualification and requirements verification, and cross-functional engineering work.

Education

Toronto Metropolitan University

Sep. 2024 – Apr. 2029

Bachelor of Engineering (B.Eng), Electrical Engineering – GPA: 4.13/4.33

Toronto, Ontario

- Progressing toward a hardware-focused electrical engineering degree with coursework spanning analog circuits, digital logic, data structures, object-oriented programming, and scientific computing.
- Available for internship or co-op work with multiple academic terms remaining after placement completion.

Technical Skills

Languages: Python, C, C++, Java, SystemVerilog/VHDL, MATLAB, Bash, SQL

Hardware & Circuit Tools: Quartus II, ModelSim, Icarus Verilog, Multisim, KiCad/SPICE simulation, ESP32, Oscilloscope, Function Generator, Multimeter, Bench Power Supply, Soldering

Verification & Data Tools: cocotb, pytest, JUnit, Reference/Golden Modeling, Constrained-Random Testing, Pandas, NumPy, Matplotlib, PyTorch, GNUplot, Git/GitHub

Technical Areas: RTL design, functional verification, FSM and ALU architecture, BJT/MOSFET biasing, rectifiers, voltage regulation, amplifier analysis, motor drive (H-bridge, PWM), embedded firmware, requirements-based verification, algorithm analysis

Documentation & Collaboration: Lab reports, design justification, technical presentations, structured debugging, team handoff, stakeholder communication

Relevant Coursework

Hardware and Circuit Design | *ELE404 Electronic Circuits I, COE328 Digital Systems*

2025 – 2026

- Electronic Circuits I: diode models, rectifiers, voltage regulation, BJT and MOSFET operating regions, small-signal models, common-emitter/common-collector amplifier analysis, multi-stage buffering, distortion, gain, input/output resistance, and comparison of theoretical, simulated, and measured hardware behavior.
- Digital Systems: Boolean algebra, Karnaugh Maps, NAND-only implementation, combinational logic, multiplexers, encoders, decoders, Johnson counters, sequential circuits, finite state machines, VHDL simulation, and FPGA-style logic verification relevant to controller design.

Algorithms, Programming, and Software Systems | *COE428 Data Structures and Algorithms, COE318 Object-Oriented Programming*

- Data Structures and Algorithms: recursion, insertion sort, merge sort, asymptotic analysis, stacks, heaps, state machines, structured input parsing, modular C design, command-line debugging, and implementation against formal specifications.
- Object-Oriented Programming: abstraction, encapsulation, inheritance, polymorphism, interfaces, method overriding/overloading, JUnit-based testing, exception handling, validation logic, and modular Java system design.

Scientific Computing and Data Analysis | *CPS188 Computer Science for Engineers, Independent Python/ML Work*

2024 – 2025

- Computer Science for Engineers: C fundamentals, pointers, arrays, file I/O, structures, binary files, recursion, data visualization using GNUplot, modular programming, debugging, and scientific reporting through lab and project work.
- Independent technical study: Python-based machine learning, time-series preprocessing, LSTM model training in PyTorch, quantitative evaluation, and technical visualization for data-driven engineering workflows.

Technical Projects

RV32I RISC-V Core – RTL & Design Verification | *SystemVerilog, Python, cocotb, pytest* 2026

- Verified three RV32I datapath blocks to a self-checking standard by building a **cocotb** and **pytest** harness that checks each block against an independent **Python** reference model across **12,000+** directed and constrained-random vectors.
- Designed synthesizable **SystemVerilog** RTL for a **10-operation** ALU, a **32x32-bit** register file with hardwired x0 and synchronous write, and an immediate generator covering all **5** RV32I formats (I, S, B, U, J).
- Isolated single-bit miswiring in the B- and J-type immediate paths that random stimulus alone missed, by writing a **walking-ones** bit-sweep testbench that exercises every immediate wire individually.
- Cut repeat-failure debug time by making every regression run seeded and replayable and by tracing mismatches at the **waveform level** to the exact failing signal.

TakeoffBot – Six-Motor Embedded Drive Platform | *ESP32, C++, H-Bridge Drivers* Summer 2026

- Brought a **six-motor** embedded platform from parts to a working system on the bench, by writing **C++** firmware for **LEDC PWM** speed control, H-bridge motor drive, and servo actuation on an **ESP32**.
- Eliminated brownout resets crashing the controller, by diagnosing a power-domain fault and isolating the **7.4 V** motor supply from logic power.
- Cut false obstacle stops from sensor noise, by applying a **median filter** to the **HC-SR04** ultrasonic input at a **20 cm** stop threshold, and added a dead-man timeout that halts all drive output within one control cycle of a lost connection.

MOSFET Amplifier Design Project | *SPICE, GlobalFoundries 180nm CMOS* Winter 2026

- Met a full open-loop amplifier spec (**60 dB** gain, **500 kHz** bandwidth, **100 k Ω** input resistance, **1.5 V_{pp}** swing, under **10%** gain loss under load) on a **3.3 V** single supply, by selecting a cascaded NMOS/PMOS topology and validating it in **SPICE**.
- Held total DC power under **1 mW** and branch current under **200 μ A**, by sizing and biasing every transistor from hand calculations at **0.5 μ m** channel length and refining operating points through DC sweeps.
- Reconciled predicted and simulated gain, bandwidth, swing, and power under a **10 k Ω / 2 pF** load, by documenting each discrepancy and trade-off in a formal design report.

Diode, Regulator, and Rectifier Laboratory Portfolio | *Oscilloscope, Multimeter, Function Generator, Multisim* Winter 2026

- Characterized the **1N4148** diode and extracted its ideality factor and saturation current, by measuring forward-bias voltage-current behavior and fitting measured against simulated data.
- Ranked **3** voltage regulator topologies on load regulation, by sweeping load current through a **2N3904**-based adjustable load across resistor-divider, diode-drop, and **1N4735** Zener designs.
- Quantified DC output and ripple for a full-wave bridge rectifier, by deriving its transfer characteristic and testing smoothing-capacitor, load-variation, and source-resistance effects on the bench.
- Produced review-ready lab documentation with calculations, plots, and error analysis comparing theoretical, simulated, and measured results and identifying nonideal component effects.

BJT Amplifier and Buffering Laboratory Portfolio | *2N3904, Small-Signal Analysis, Bench Verification* Winter 2026

- Predicted and verified gain and input/output resistance for common-emitter and common-collector amplifiers, by biasing **2N3904** stages in active mode and deriving g_m , r_e , and r_π for both DC and small-signal analysis.
- Measured amplifier input and output resistance on the bench, by applying a practical two-port method and evaluating distortion, maximum input swing, and emitter-bypass effects on gain and linearity.
- Demonstrated a buffered **two-stage CE-CC** amplifier isolating a gain stage from heavy loading, by constructing the emitter follower and comparing loaded source-to-output performance against hand calculations.

Digital Systems and 8-bit ALU Design Portfolio | *VHDL, Quartus II, ModelSim, DE10-Lite FPGA* Fall 2025

- Built a modular **8-bit** ALU supporting ADD, SUB, AND, OR, and XOR, by organizing a control unit, register block, core ALU logic, and shared bus in **VHDL**.
- Verified functional correctness across all operation paths, by running **ModelSim** waveform simulations and tracing signals to isolate mismatches between expected and simulated behavior.
- Implemented Mealy and Moore finite state machines to sequence display outputs with explicit invalid-state handling, and reduced gate count on selected blocks through Karnaugh-map and NAND-only implementation on the **DE10-Lite**.

TMU Robotics Rover Team – ROS 2 Drive Stack | *ROS 2 Humble, Ubuntu 22.04, Linux* 2026

- Brought a six-wheeled rover's drive system online, by debugging a chain of dependency and launch-file misconfigurations in a **ROS 2 Humble** stack on **Ubuntu 22.04**.
- Resolved integration issues between software packages and hardware interfaces in a **Linux** environment, by collaborating with rover team members through iterative testing.

Ice Concentration Analysis on the Great Lakes | *C, GNUplot, Scientific Data Processing* 2024 – 2025

- Processed NOAA daily ice-concentration data for **6** Great Lakes across **2021–2024**, by building a modular C program using **2D arrays** and structured functions to compute averages, extrema, peak dates, and year-over-year trends.
- Automated per-lake and comparative trend visualization, by generating GNUplot scripts from the processed datasets, and supported reproducibility through clear commenting and modular code handoff in a team setting.

Algorithms and Systems Programming Portfolio | *C, GCC, make, Command-Line Testing*

Winter 2026

- Measured comparison, copy, and swap counts for insertion sort and merge sort behind a common interface, by instrumenting metrics and analyzing best-, average-, and worst-case behavior against derived equations.
- Built a finite state machine simulator supporting transition updates, reachability analysis, and unreachable-state deletion under strict I/O requirements, plus a recursive Towers of Hanoi solver separating functional output from debug traces.
- Validated simplified XML with a stack-based parser and ordered integers through a heap, by generating structured tree output and sorted and reverse-sorted results from abstract data structures.

Object-Oriented Engineering Software Portfolio | *Java, JUnit, Modular Design*

Fall 2025

- Modeled DC circuits, vehicle rental, and parcel shipping in Java, by applying inheritance, interfaces, and polymorphism across testable class hierarchies with JUnit validation and exception handling.
- Linked object-oriented design to electrical-system representation, by implementing **Resistor** and **VoltageSource** classes with SPICE-style output formatting and structured text generation.

Stock Price Prediction Using LSTM | *Python, PyTorch, Pandas, Matplotlib*

2026

- Predicted next-day AAPL closing prices, by building an end-to-end **PyTorch** LSTM pipeline on **5+ years** of market data with a **30-day** sliding window and a chronological **80/20** split to prevent leakage.
- Evaluated model accuracy with **RMSE** after inverse scaling, by training with MSE loss, Adam optimization, and backpropagation through time and visualizing predicted versus actual error trends in Matplotlib.

Professional Experience

Awksion (Startup)

Jan. 2026 – Apr. 2026

Software Engineering Intern

Remote

- Ingested and validated **2,341** North American venues into a **SQLite/WAL** database, by architecting an idempotent **Python** ETL pipeline orchestrating **Selenium** scrapers and the **OSM Overpass API** as independently re-runnable stages.
- Backfilled missing capacity data across **350+** venues, by building a **scikit-learn** estimation model fusing geospatial footprint analysis with fire-code occupancy heuristics and multi-tenant building disambiguation.
- Sustained stable multi-province extraction over long unattended runs, by implementing **100-page / 60-min** driver cycling, **WebDriverWait** condition logic, and append-safe writes that prevent data loss on failure.
- Scored venue-artist leads on city, capacity, and genre overlap, by tiering **244** artists into **10k–100k** follower buckets through the Last.fm and Spotify APIs behind a 3-layer filter, and communicated trade-offs to a non-technical founder in regular meetings.

Toronto Metropolitan University – Student Life and Learning Support (SLLS) Sep. 2025 – Apr. 2026

Math Support Assistant

Toronto, Ontario

- Supported **30–50** students per shift, by managing real-time scheduling and triaging session requests on the SLLS platform in a busy service environment.
- Maintained **100+** attendance, payroll, and inventory records with reporting accuracy, by standardizing tracking across Excel and Google Sheets for weekly operational reports.
- Reduced missed sessions and idle tutor time, by resolving **10+** scheduling conflicts per shift through coordinated outreach with students and faculty.

HUNGRY

Sep. 2025 – Present

Catering Delivery Captain

Toronto, Ontario

- Coordinated logistics for high-volume, multi-stop catering orders across Toronto, by verifying delivery details and order accuracy across clients, vendors, and internal team members.
- Kept multi-stop pickups and drops on schedule under time pressure, reinforcing reliability and attention to detail in fast-paced operations.

Rogers Centre

Sep. 2025 – Nov. 2025

Vending Attendant

Toronto, Ontario

- Maintained inventory and point-of-sale accuracy under event-time pressure, by tracking stock movement and handling reconciliation and restocking across major events.

Leadership / Extracurricular

Engineering Outreach, Toronto Metropolitan University

2024 – Present

Ambassador

Toronto, Ontario

- Led hands-on STEM workshops for students in Grades **1–12** across **5+** programs (Badge Day, Go CODE Girl, Go ENG, CAGIS, FuturEng), by guiding LED circuit, Arduino, and glider builds.
- Built public speaking and technical communication skills, by translating engineering concepts for younger learners and parents.

GVIC Engineering Conference, Toronto Metropolitan University

2025 – 2026

Organizer

Toronto, Ontario

- Helped organize a student engineering conference, by coordinating schedules, workshops, and follow-up action items across the planning team.

MEC 2025, Toronto Metropolitan University

May 2025 – Nov. 2025

Organizer

Toronto, Ontario

- Co-developed a debate-style event on engineering ethics, and supported planning and execution for a student-facing competition.